PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Part-1 Sprint Backlog and Implementation Start

Hostel Management System (HMS)

Team 3 | 07 October 2026

Project Team

Name	SRN
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Revision History

Version	Date	Contributor	Revision / Contribution
1.0	05-10-2026	G.V.N. Sreeharsha	Created initial Part-1 sprint goal and backlog structure.
1.1	07-10-2026	Chennamraju Vaishnavi	Added testing tasks, execution activities and acceptance criteria.
1.2	07-10-2026	Devansh Gupta	Added architecture, component and allocation/payment implementation tasks; finalised acceptance criteria.

Sprint Goal

Complete the Part-1 documentation (SRS, SAD, Test Plan) and set up the Django + MySQL project foundation so that implementation of the modules C2 to C8 can begin.

Work Distribution

Member	Team Role	Documentation	Implementation	Testing
G.V.N. Sreeharsha	Team Lead / Product Owner	SRS, RTM, final integration review	C2 Auth & Access, C8 Audit	TC-AUTH, TC-USR, TC-SEC
Chennamraju Vaishnavi	QA / Test Lead	Complete Test Plan	C3 Student, C4 Hostel/Room/Bed	TC-STU, TC-HST, TC-QUAL
Devansh Gupta	Architecture / Development Lead	SAD architecture + design, component UML, SD-2	C5 Allocation, C6 Payment, C7 Reports	TC-ALC, TC-PAY, TC-RPT, TC-PERF, TC-OPS

Sprint Backlog

ID	Task	Owner	Priority	Related requirements	Status
B-01	Create GitHub repository, Django project and app skeletons (accounts, students, hostels, allocations, payments, reports, audit)	Sreeharsha	High	Constraints (SRS 2.5)	Planned

B-02	Set up MySQL 8.0 database and shared models (C9) with the one-active-allocation constraints	Devansh	High	BR-1, BR-2, NFR-5	Planned
B-03	Login, logout, lockout and session timeout	Sreeharsha	High	REQ-1 to REQ-3, NFR-9, NFR-12	Planned
B-04	Role-based access control and user administration	Sreeharsha	High	REQ-4, NFR-7, NFR-8	Planned
B-05	Audit logging	Sreeharsha	Medium	NFR-11	Planned
B-06	Student registration, search, update and deactivate	Vaishnavi	High	REQ-5 to REQ-9	Planned
B-07	Hostel, room and automatic bed creation; available-beds list	Vaishnavi	High	REQ-10 to REQ-13	Planned
B-08	Allocate, transfer and release with atomic transactions and allocation history	Devansh	High	REQ-14 to REQ-19	Planned
B-09	Payment recording, receipt PDF and payment status	Devansh	Medium	REQ-20 to REQ-22	Planned
B-10	Occupancy report with CSV export	Devansh	Medium	REQ-23	Planned
B-11	Student My Allocation view	Vaishnavi	Medium	REQ-24	Planned
B-12	Test environment and test data (5,000 students, 2,000 beds); Django unit tests for each module	Vaishnavi	High	Test Plan sections 5 and 6	Planned
B-13	Security, performance and backup testing	Sreeharsha	Medium	NFR-1 to NFR-4, NFR-6, NFR-10	Planned

Part-1 Deliverables

- Software Requirements Specification (SRS)
- Software Architecture and Design Specification (SAD)
- Software Test Plan
- Component UML Diagram (CD-01)
- UML Use Case Diagram (UC-01)
- Two UML Sequence Diagrams (SD-01 Student Registration, SD-02 Bed Allocation)
- Requirement Traceability Matrix (SRS Appendix C and Test Plan 13.2)
- 33 test cases defined (21 functional, 8 security, 3 performance, 1 other non-functional)

Definition of Done

- All three members have contributed to every document, as recorded in each document’s revision history.
- SRS, SAD and Test Plan are mutually consistent in requirement IDs (REQ-1 to REQ-24, NFR-1 to NFR-12).
- Every functional, non-functional and security requirement has test and traceability coverage.
- Architecture components and design interfaces match SRS terminology.
- UML diagrams use the same actors, components and flows as the SRS and SAD.
- Implementation modules are assigned and can be integrated in the sprint.
- Repository contains the Part-1 documents and editable diagram sources.

Product Ownership

Product ownership is assigned to G.V.N. Sreeharsha (Team Lead / Product Owner) for requirement prioritisation, backlog decisions, scope control and final review. This does not replace the technical and documentation responsibilities of Vaishnavi (QA / Test Lead) and Devansh (Architecture / Development Lead).